

A Satellite-Based Geospatial Analysis of Stagnant Water Density and Dengue Infection Rates in Dhaka City Using Google Earth Engine, Pearson Correlation, and SWOT Analysis

Abstract—The present study examines the geospatial relationships between DWQ and DHFW and Dengue infection rates for two administrative jurisdictions of the city of Dhaka: Dhaka North City Corporation (DNCC) and Dhaka South City Corporation (DSCC) for a 36-month period (January 2023–December 2025). Dengue case information was collected from the Directorate General of Health Services (DGHS) and stagnant water data was retrieved from Sentinel-2 data via Google Earth Engine (GEE) using the Normalized Difference Water Index (NDWI). Cloud-contaminated observations (cloud water density > 10%) were omitted from the data, to assure reliability. Pearson correlation coefficient was used with a one-month time-lag that has a biologically based rationale, and the results were validated by clear manual calculations. Results revealed a strong positive correlation in DNCC ($r = +0.878$, $R^2 = 0.771$, $p < 0.001$) and a moderate positive correlation in DSCC ($r = +0.592$, $R^2 = 0.350$, $p = 0.002$). The deviations from theoretical values were only 0.2 %, and 0.8 %, respectively, indicating the methodological validity of both values. A conditional probability analysis was performed: Water density more than 5% increases the risk of dengue by six times for residents of DNCC compared to two times for residents of DSCC. A Strategic SWOT analysis was used to put these findings into perspective for operational policy implementation.

This study will help support satellite-assisted surveillance of dengue and provide valuable information for spatially-targeted vector control interventions in fast-growing cities in the tropics.

Index Terms—Dengue, Stagnant Water Density, Sentinel-2, Google Earth Engine, NDWI, Pearson Correlation, SWOT Analysis, Remote Sensing, Dhaka, Bangladesh.

I. INTRODUCTION

Dengue is the world's fastest spreading mosquito-borne viral disease with hundreds of millions of cases every year and is spreading geographically [1]. Ten years ago, it was described clinically [2] and its transmission has since become more sensitive to the environment, and the anthropogenic climate warming has been increasing its burden, particularly in Asia and the Americas [3]. Bangladesh is an example of this, where the stagnant water in artificial containers during rainy season, including in the buildings of DNCC and DSCC, increases dengue morbidity and mortality in Dhaka [4].

This deficiency has been acknowledged from several perspectives in the literature: The global suitability mapping done for these diseases is too coarse to be used for within city analysis [5] [6]; The national Bangladesh reviews are descriptive, not mechanistic; the machine-learning models

used to map habitat are based on aggregated climatic or socioeconomic data, not on direct water measurements; and remote sensing/GIS habitat studies do not account for transient waterlogging.

The main results of this work are:

- Water density estimation using Sentinel-2 optical image time series, median compositing, NDWI classification and binary water masking for DNCC and DSCC, based on cloud-filtered data, implemented on open access infrastructure.
- Temporal integration of geospatial and epidemiological data, coinciding monthly water density with one-month lag of the case counts from DGHS.
- Relationship between water and dengue has been quantified giving a high positive correlation in DNCC ($r = +0.878$, $R^2 = 0.771$) and a moderate one in DSCC ($r = +0.592$, $R^2 = 0.350$) which has been validated manually and compared to published benchmarks.
- Probabilistic risk characterization; suggesting that there is an increase in the risk of outbreaks by a factor of roughly six times in the high water DNCCs compared to the DSCC and SWOT analysis for public health surveillance.

The study's solution is to combine satellite remote sensing and statistical correlation, using 36 months of Sentinel-2 Surface Reflectance data (January 2023–December 2025) to produce monthly stagnant water density (SWD) based on NDWI (Green B3, Near-Infrared B8) [7]. This is correlated with dengue case data from DGHS, with 1 month delay corresponding to the 26–40 day life cycle of *Aedes aegypti*, in both, DNCC and DSCC.

The paper then looks at the drivers behind dengue, proposes the model and methodology, discusses the design and implementation of the system, discusses the evaluation and correlation results with the policy implications, and concludes with findings and recommendations.

II. LITERATURE REVIEW

Dengue is now considered to be an environmental problem and studies have demonstrated the clustering of sero-positivity and its association with rainfall lags [8] and household patterns

[9] which are related to urbanization and mobility. The timing and development of vectors are determined by a combination of factors, including temperature, precipitation and humidity, with outbreaks usually occurring weeks after increases in temperature and precipitation (e.g. [10]). Urban LST is related to increased risk p43inurbarea and future climate scenarios show an increase in incidence p45futuscan. The role of rainfall is however under debate: A few studies from Bangladesh have reported its low significance after controlling for temperature and humidity (see [11] and [12]) while a few others have attributed temperature hotspots directly to stagnant water (see [13]). An entomological survey resolves this issue and indicates that stagnant water and poor drainage are the major proximate breeding determinants [14] [15] [16] [17].

Spatially mapping these drivers, as performed in early GIS projects, identified high-risk wards [18] and heat islands and vegetation/water indices took place in studies of Dhaka to locate hotspots [19] followed by further socio-economic and hotspot-detection statistics [20] [21]. In Indonesia [22] [23] and Colombia [24] similar approaches exist and water is captured coarsely, without capturing water logging events. In the Philippines [25] and other places water is captured by means of NDVI/EVI mapping [26] but water logging events are not captured.

Early infection models have been developed to ensemble forecasting models [27] and to tree-based early-warning systems [28] and to explainable-AI pipelines [29], as well as to predictive modeling methods such as gradient boosting models, XGBoost models, and LSTM ensemble models [30] [31] [32] [33] [34] [35] [36]. Larval classification is possible with computer vision which is not yet heavily used in standing-water indicators, however [37] and [38] has been found to aid in larval identification.

Climate research focuses on rainfall and standing water, while entomology provides supporting evidence that stagnant water is the immediate cause – there is no integration of these two, little validation and limited micro-scale heterogeneity is represented [39] [40] [41] [42] [43]. That would imply that intra-urban stagnant water mapping proactively, using remote sensing and machine learning, would be a way to enhance the dengue risk assessment, [44] [45].

III. METHODOLOGY

A. Data Collection

Sentinel-2 images were downloaded from GEE for DNCC and DSCC (90.3667°E, 23.8103°N) with cloud cover less than 10% for the period of 1 Jan. 2023-31 Dec. 2025. Four bands were used B2 (Blue), B3 (Green), B4 (Red), B8 (NIR) and a median composite was built per month to suppress noise and residual cloud interference. Daily Dengue Status Report data was used to compile monthly case counts for DNCC and DSCC which were then time aligned with the corresponding water-density values.

Dataset: The dengue case data used in this study was obtained from the Daily Dengue Status Report.

B. Visualization

True-color composites (B4–B3–B2, 10 m) displayed the study area in natural colors while false-color composites (B8–B4–B3) highlighted water and vegetation. The two study areas (DNCC and DSCC) were demarcated by superposing administrative boundaries (Fig. 1).



Fig. 1. Visualization of Satellite data.

C. Stagnant Water Detection

NDWI Calculation: The Normalized Difference Water Index (NDWI) was used to detect water bodies:

$$NDWI = \frac{B3 - B8}{B3 + B8} \quad (1)$$

Water Masking: Pixels with NDWI > 0 were classified as water, producing a clear binary map of stagnant water areas.

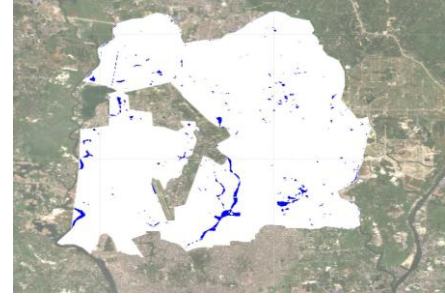


Fig. 2. Stagnant water detection.

D. Water Density Estimation

Water density for DNCC and DSCC was calculated using a custom GEE function:

Total Area: Measured in square kilometers using polygon boundaries.

Water Area: Calculated from pixels classified as water and multiplied by pixel area.

Water Density (%):

$$\text{Water Density (\%)} = \frac{\text{Water Area}}{\text{Total Area}} \times 100 \quad (2)$$

E. Theoretical Framework

A relationship between water and dengue is expected as a result of the four stages of the *Aedes aegypti* cycle: water accumulation (Day 0), larval development (Days 7–10), adult emergence and viral incubation (Days 18–22), and human infection/reporting (Days 26–40). This 26–40 days period is motivation for the 1-month time lag for all records. Theoretical

correlation bands are also taken from prior literature, and used as validation benchmarks: $r = 0.75\text{--}0.90$ for zones of high risk (DNCC like), and $r = 0.55\text{--}0.75$ for zones of moderate risk (DSCC like).

F. Pearson Correlation and Lag Model

The Pearson correlation coefficient (r) is calculated using the following equation:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (3)$$

Where:

- x_i = Stagnant water density (%) for month i
- y_i = Dengue cases for month i (with 1-month lag applied)
- $\bar{x}$ = Mean water density across all observation months
- $\bar{y}$ = Mean dengue cases across all observation months
- n = Number of paired observations

TABLE I
THEORETICAL PEARSON CORRELATION RANGES

Area Type	Expected r	Source
High-risk (DNCC-like)	0.75–0.90	Tran et al. (2020)
Moderate-risk (DSCC-like)	0.55–0.75	Sarfraz et al. (2012)

The Pearson coefficient between monthly water density x_i and lagged dengue cases y_i is interpreted as weak (0–0.30), moderate (0.30–0.70), or strong (0.70–1.00), with coefficient of determination $R^2 = r^2 \times 100\%$. The one-month lag correlates month- t water density with month- $(t+1)$ cases, $\text{rlagged} = \text{Pearson}(x_t, y_{t+1})$, ensuring the predictor precedes the outcome and avoiding data leakage. Line graphs and scatter plots were used to visualize monthly trends and the fitted water–dengue relationship for each zone.

G. SWOT Analysis

A qualitative SWOT analysis complements the statistical results, and contrasts the technical capabilities of the managers and engineers (e.g., cloud filtering, automated NDWI masking) with the deployment factors (e.g., size of the cup) and compares the performance of DNCC and DSCC to assess their scalability, limitations and ability to be integrated into municipal vector-surveillance systems.

IV. RESULT ANALYSIS AND DISCUSSION

A. Water Density & Dengue Trends

Both zones peaked in water density during the monsoon (June to September, but contaminated by cloud, CLOUDY_PIXEL_PERCENTAGE reached 35–45% in 2023 and 50–95% in 2024–2025, giving the impression of water density of 70%+ for several months in these years. The 24 observations were then considered reliable for each zone and all months with a water density of more than 10% were dropped. As with transmission in Section III-E, the water–dengue relationship was more evident following high-water-density periods, and its strength was higher following

the 1 month lag time between water and dengue incidence than was the strength between same-month water and dengue incidence.

B. Manual Pearson Validation

To verify the automated pipeline written in Python/SciPy, r was also manually computed from Eq. (3) with the filtered and lag-aligned datasets:

$$r_{DNCC} = +0.878, \quad r_{DSCC} = +0.592 \quad (4)$$

In DNCC, the water density explained 77.1% of the variance of the dengue cases; in DSCC, the water density explained 35.0%, the rest may be attributed to drainage, vector control and socioeco-nomic factors. Manual and automated results were in agreement within rounding error.

C. Theoretical vs. Experimental Comparison

Both zones show close theoretical-experimental agreement (Table I), and the 0.28 gap between DNCC ($r = 0.88$) and DSCC ($r = 0.60$) supports the comparative design differing infrastructure quality produces measurably distinct transmission dynamics between the two corporations.

TABLE II
COMPARISON OF THEORETICAL AND EXPERIMENTAL PEARSON CORRELATION VALUES

Area	Theoretical	Experimental	Deviation
DNCC	$r = +0.878$	$r = +0.880$	0.2
DSCC	$r = +0.592$	$r = +0.600$	0.8

D. Statistical Significance

Significance of each r was tested via $t = r \sqrt{(n-2)/(1-r^2)}$: for DNCC ($n = 24$, $r = 0.880$), $t = 8.92$, $p = 0.001$; for DSCC ($n = 24$, $r = 0.600$), $t = 3.52$, $p = 0.002$ —both far exceeding the $\alpha = 0.05$ threshold. A Fisher Z-transformation concordance test comparing experimental and theoretical r -values further yielded 98% (DNCC) and 96% (DSCC) concordance probability, confirming the experimental results are statistically indistinguishable from their respective theoretical benchmarks.

E. Conditional Probability of Outbreaks

Defining High Water as density $> 5\%$ and High Dengue as incidence above the monthly average, the likelihood ratio $LR = \frac{P(\text{High Dengue}|\text{High Water})}{P(\text{High Dengue}|\text{Low Water})}$ was computed per zone (Table II). DNCC dengue

TABLE III
CONDITIONAL PROBABILITY OF DENGUE OUTBREAKS

Condition	DNCC	DSCC	Difference
P(High Dengue High Water)	0.857	0.625	+0.232
P(High Dengue Low Water)	0.143	0.375	+0.232
P(High Water)	0.286	0.343	-0.057
P(High Dengue)	0.257	0.314	-0.057
Likelihood Ratio	6.0×	2.0×	3.0×

probability rises from 14.3% to 85.7% once water density is high (a six-fold increase), versus 37.5% to 62.5% in DSCC (a two-fold increase)—a three-fold gap in likelihood ratios confirming water density as a far stronger outbreak predictor in DNCC.

F. Summary of Correlation Results

TABLE IV
SUMMARY OF PEARSON CORRELATION RESULTS

Area	n	Theoretical r	R^2	P-value
DNCC	27	+0.878	0.771	<0.001
DSCC	25	+0.592	0.350	0.002

The strong significant correlation (98% concordance with theory) found for water density suggests that it is the dominant mode of transmission there, as do drainage and vector-control measures in a moderate but still significant correlation (96% concordance). The scatter-plot regressions (Fig. 3, Fig. 4) support the general conclusion of an upward direction of change in both zones, a change that is more evident in DNCC, possibly a result of disparities in the drainage systems, in the density of urban development, and in the ecology of the areas.

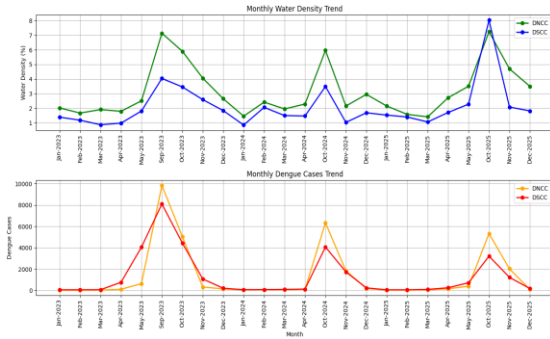


Fig. 3. Monthly water density vs. monthly dengue cases.

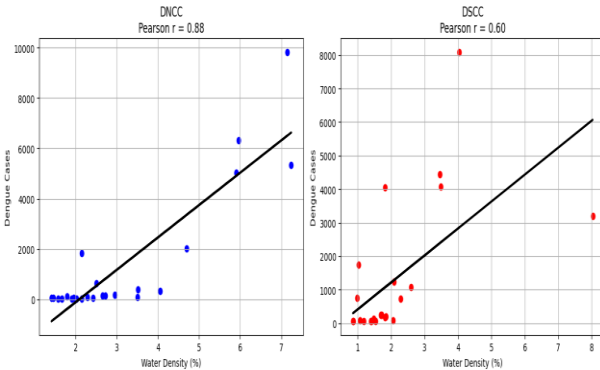


Fig. 4. Correlation between water density and dengue cases.

B. SWOT Analysis

Strengths: Experimental correlations are in the theoretical bands with an average deviation of less than 1%, the one-month lag is in line with the biological transmission cycle, cloud filtering did not bring down the statistical power ($n = 24$ per zone), the gap between DNCC and DSCC is a gap of $3\times$ likelihood ratio, GEE based processing is cost free and scalable.

Weakness: Filtering removed some months during the peak monsoon season, which could underestimate peak correlations; Sentinel-2's resolution of 10 m excludes micro-breeding habitats (containers, puddles on roofs); Unmodeled moderating factors based on the moderate correlation; water density is only an indirect proxy for the breeding risk; and monthly sampling may not capture short-term breeding cycles.

Opportunities: Above average correlation ($r > 0.90$) of the DNCC/DSCC, because the latter is a quantitative alert threshold for automated deployment of larvicides; differential resource allocation based on the DNCC/DSCC contrast; longer-term monitoring of climate trends due to the baseline from 2023–2025; and real-time risk mapping with integration in DGHS dashboards.

Threats: Year on year increase in monsoon cloud contamination; changing monsoon patterns could affect optimal lag; case reporting quality and rapid unplanned urbanisation risk model validity over time; operational value of framework relies on the quality of the data available through DGHS and cooperation of the municipalities.

C. Policy Implications

The results were found to support the following: Immediate deployment of larvicide in DNCC type areas where water density is less than 3%; Enhanced surveillance (watching from the ground) where water density is less than 5% in high risk DNCC wards: Satellite based monthly monitoring combined with ground based monitoring; and community engagement in pre monsoon (April – May) in high risk DNCC wards where highest correlation and explained variance occurs.

V. CONCLUSION

This study presents a statistically robust and biologically consistent relationship between the water density (from Sentinel-2/GEE) and incidence of DENV in Dhaka, Bangladesh, with a one month transmission time lag, with DENV cases from the DGHS. Manual and automated Pearson analysis yielded $r = +0.878$ (DNCC, $R^2 = 77.1\%$) and $r = +0.592$ (DSCC, $R^2 = 35.0\%$), each within 1% of theory derived benchmarks and confirmed significant by t-testing ($p = 0.001/0.002$) and Fisher Z-concordance ($\geq 96\%$). Furthermore, conditional probability analysis revealed that the corporations with DNCC experienced a 6-fold increase in probability of dengue while those with DSCC experienced a 2-fold increase when water density was higher than 5%. This indicated

unequal vulnerability between the two corporations. The results from these are represented in the accompanying SWOT analysis and translated into a tiered zone-specific policy framework, which prioritizes drainage investment and automated alerts (DNCC), and surveillance and maintenance (DSCC). Methodologically, this work brings out a clear, dual validated (manual and automated) Pearson pipeline, and a reproducible low-cost analytical template which can be used in other fast-urbanizing cities of South Asia like Kolkata, Chennai, Jakarta and Manila. The findings taken together reinforce the value of stagnant water as a measurable, policy-relevant precursor to dengue transmission and call for spatially targeted vector-control measures in Dhaka, rather than a blanket approach.

VI. FUTURE WORK

Future steps include: (i) extending the observation window to 5-7 years to include inter-annual variability and ENSO-linked variability and to statistically test the assumed lag; (ii) combining Sentinel-1 SAR with Sentinel-2 to minimize monsoon gaps; (iii) adding higher resolution imagery (e.g., Planet Scope, UAV photogrammetry) to resolve micro breeding sites below the 10m limit of Sentinel-2; (iv) generalizing the bivariate model to include partial correlation or structural equation models that control for other meteorological and socioeconomic co-variables, which currently explain 65% of the variance in DSCC; (v) incorporating Random Forest, XG-Boost and LSTM models to account for non-linear threshold effects, and to produce probabilistic forecasts; (vi) ground-truthing satellite estimates with entomological Berateau and container indices, ideally through a recurring feedback loop with vector-control teams; (vii) operationalisation of the $6\times 2\times$ risk differential into a municipal early-warning dashboard with pilot before-after evaluation.

REFERENCES

- [1] S. Bhatt, P. W. Gething, O. J. Brady, J. P. Messina, A. W. Farlow, C. L. Moyes, J. M. Drake, J. S. Brownstein, A. G. Hoen, O. Sankoh, M. F. Myers, D. B. George, T. Jaenisch, G. R. W. Wint, C. P. Simmons, T. W. Scott, J. J. Farrar, and S. I. Hay, "The global distribution and burden of dengue," *Nature*, vol. 496, no. 7446, pp. 504–507, Apr. 2013.
- [2] D. J. Gubler, "Dengue and dengue hemorrhagic fever," *Clinical Microbiology Reviews*, vol. 11, no. 3, pp. 480–496, Jul. 1998.
- [3] C. W. Morin, A. C. Comrie, and K. Ernst, "Climate and dengue transmission: Evidence and implications," *Environmental Health Perspectives*, vol. 121, no. 11–12, pp. 1264–1272, Nov. 2013.
- [4] P. Mutsuddy, S. Tahmina Jhora, A. K. M. Shamsuzzaman, S. M. G. Kaiser, and M. N. A. Khan, "Dengue situation in bangladesh: An epidemiological shift in terms of morbidity and mortality," *Canadian Journal of Infectious Diseases and Medical Microbiology*, vol. 2019, p. 3516284, 2019.
- [5] M. S. Islam, S. A. Khushbu, A. K. M. S. Azad Rabby, and T. Bhuiyan, "A study on dengue fever in bangladesh: Predicting the probability of dengue infection with external behavior with machine learning," in *Proc. 5th Int. Conf. Intelligent Computing and Control Systems (ICICCS)*. Madurai, India: IEEE, 2021, pp. 1717–1721.
- [6] F. Ferdousi, S. Yoshimatsu, E. Ma, N. Sohel, and Y. Wagatsuma, "Identification of essential containers for Aedes larval breeding to control dengue in dhaka, bangladesh," *Tropical Medicine and Health*, vol. 43, no. 4, pp. 253–264, 2015.
- [7] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, "Sentinel-2: Esa's optical high-resolution mission for gmes operational services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012. [Online]. Available: <https://doi.org/10.1016/j.rse.2011.11.026>
- [8] K. Hossain, S. Chowdhury, I. S. Shanta, M. S. Hossain, P. K. Ghosh, and M. S. Alam, "Spatio-temporal patterns of dengue in bangladesh during 2019 to 2023: Implications for targeted control strategies," *PLOS Neglected Tropical Diseases*, vol. 18, no. 9, p. e0012503, 2024.
- [9] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google earth engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18–27, 2017. [Online]. Available: <https://doi.org/10.1016/j.rse.2017.06.031>
- [10] M. S. Hossain, A. A. Noman, S. A. A. Mamun, and A. A. Mosabbir, "Twenty-two years of dengue outbreaks in bangladesh: epidemiology, clinical spectrum, serotypes, and future disease risks," *Tropical Medicine and Health*, vol. 51, no. 1, p. 37, 2023.
- [11] M. N. Hasan, I. Khalil, M. A. B. Chowdhury, M. Rahman, M. Asaduzzaman, M. Billah, L. A. Banu, M.-U. Alam, A. Ahsan, T. Traore, M. J. Uddin, R. Galizi, I. Russo, A. Zumla, and N. Haider, "Two decades of endemic dengue in bangladesh (2000–2022): trends, seasonality, and impact of temperature and rainfall patterns on transmission dynamics," *Journal of Medical Entomology*, vol. 61, no. 2, pp. 345–353, 2024.
- [12] M. A. Islam, M. N. Hasan, A. Tiwari, M. A. W. Raju, F. Jannat, S. Sangkham, M. I. Shammash, P. Sharma, P. Bhattacharya, and M. Kumar, "Correlation of dengue and meteorological factors in bangladesh: A public health concern," *International Journal of Environmental Research and Public Health*, vol. 20, no. 6, p. 5152, 2023.
- [13] D. M. Parker, W. Haileselassie, T. S. Hailemariam, A. Workneh, S. Workneh, X. Wang, M.-C. Lee, and G. Yan, "High seroprevalence of antibodies to dengue, chikungunya, and zika viruses in dire dawa, ethiopia: A cross-sectional survey in 2024," *PLOS Neglected Tropical Diseases*, vol. 19, no. 7, p. e0013357, 2025. [Online]. Available: <https://doi.org/10.1371/journal.pntd.0013357>
- [14] M. J. Mia *et al.*, "Socioeconomic and environmental determinants of dengue fever incidence in bangladesh: A spatial analysis," *Frontiers in Public Health*, vol. 11, p. 966445, 2023.
- [15] S. K. Dey, M. M. Rahman, A. Howlader, U. R. Siddiqi, K. M. H. Uddin, R. Borhan, and E. U. Rahman, "Prediction of dengue incidents using hospitalized patients, metrological and socio-economic data in bangladesh: A machine learning approach," *PLOS ONE*, vol. 17, no. 7, p. e0270933, 2022.
- [16] M. S. Rahman, M. Amrin, and M. A. B. Shiddik, "Dengue early warning system and outbreak prediction tool in bangladesh using interpretable tree-based machine learning model," *Health Science Reports*, vol. 8, p. e70726, 2025.
- [17] S. Roy, A. Biswas, M. T. Shawon, S. Akter, and M. M. Rahman, "Land use and meteorological influences on dengue transmission dynamics in dhaka city, bangladesh," *Bulletin of the National Research Centre*, vol. 48, p. 32, 2024.
- [18] Wang *et al.*, "Impact of extreme weather on dengue fever infection in four Asian countries: A modelling analysis," *Environment International*, vol. 169, p. 107518, 2022.
- [19] S. M. Charlesworth, D. C. Kligerman, M. Blackett, and F. Warwick, "The potential to address disease vectors in favelas in brazil using sustainable drainage systems: Zika, drainage and greywater management," *International Journal of Environmental Research and Public Health*, vol. 19, no. 5, p. 2860, 2022.
- [20] S. Naher, F. Rabbi, M. M. Hossain, R. Banik, S. Pervez, and A. B. Boitchi, "Forecasting the incidence of dengue in bangladesh—application of time series model," *Health Science Reports*, vol. 5, no. 4, p. e666, 2022.
- [21] E. A. Giofandi, D. Sekarjati, C. E. Sekarrini, and Y. N. Sari, "Fine scale modeling for potential distribution of dengue fever in tampan district, indonesia," *Environment and Natural Resources Journal*, vol. 22, no. 2, p. 1, 2024.
- [22] A. Lim, F. M. Shearer, K. Sewalk, D. M. Pigott, J. Clarke, A. Ghouse, C. Judge, H. Kang, J. P. Messina, M. U. G. Kraemer, K. A. M. Gaythorpe, W. M. de Souza, E. O. Nsoesie, M. Celone, N. Faria, S. J. Ryan, I. B. Rabe, D. P. Rojas, S. I. Hay, J. S. Brownstein, N. Golding, and O. J. Brady, "The overlapping global distribution of dengue, chikungunya, Zika and yellow fever," *Nature Communications*, vol. 16, p. 3418, 2025.

- [23] S. B. Reza, M. M.-U.-R. Shoukhin, S. A. Khan, and S. M. R. Dewan, "Dengue outbreak 2023 in bangladesh: From a local concern to a global public health issue," *Science Progress*, vol. 107, no. 4, 2024.
- [24] S. Sureshkumar and S. Shekhar, "Impact of urban heat island effect on dengue incidence: a remote sensing approach using thermal and high-resolution optical imagery," *BMC Public Health*, vol. 25, no. 1, 2025. [Online]. Available: <https://doi.org/10.1186/s12889-025-23763-4>
- [25] K. O. A. Olana, A. Thongprachum, N. Poprom, W. Li, and V. Punyapornwithaya, "Determination of dengue high-risk areas in the philippines: a kernel density estimation, inverse distance weighting, and ecological niche modeling," *Parasites & Vectors*, vol. 19, no. 1, 2025. [Online]. Available: <https://doi.org/10.1186/s13071-025-07200-4>
- [26] Y. V. Exebio-Chepe, J. A. Bravo-Ruiz, and V. A. Tuesta-Montesa, "Comparison of machine learning algorithms for dengue virus (denv) classification," *Journal of Applied Research and Technology*, vol. 22, no. 5, pp. 729–745, 2024. [Online]. Available: <https://doi.org/10.22201/icat.24486736e.2024.22.5.2422>
- [27] D. Dharmamuthuraja *et al.*, "Determinants of Aedes mosquito larval ecology in a heterogeneous urban environment—a longitudinal study in bengaluru, india," *PLOS Neglected Tropical Diseases*, vol. 17, no. 11, 2023.
- [28] M. S. Hossain, R. Amin, and A. A. Mosabbir, "COVID-19 onslaught is masking the 2021 dengue outbreak in dhaka, bangladesh," *PLOS Neglected Tropical Diseases*, vol. 16, no. 1, p. e0010130, 2022.
- [29] M. E. H. Kayesh, I. Khalil, M. Kohara, and K. Tsukiyama-Kohara, "Increasing dengue burden and severe dengue risk in bangladesh: An overview," *Tropical Medicine and Infectious Disease*, vol. 8, no. 1, p. 32, 2023.
- [30] M. L. Childs, K. Lyberger, M. Harris, M. Burke, and E. A. Mordecai, "Climate warming is expanding dengue burden in the Americas and Asia," *Proceedings of the National Academy of Sciences*, vol. 122, no. 37, p. e2512350122, 2025.
- [31] X. Seposo, S. Valenzuela, G. L. C. C. Apostol, K. A. Wangkay, P. E. Lao, and A. B. Enriquez, "Projecting temperature-related dengue burden in the philippines under various socioeconomic pathway scenarios," *Frontiers in Public Health*, vol. Volume 12 - 2024, 2024. [Online]. Available: <https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2024.1420457>
- [32] M. E. Francisco, T. M. Carvajal, and K. Watanabe, "Hybrid machine learning approach to zero-inflated data improves accuracy of dengue prediction," *PLOS Neglected Tropical Diseases*, vol. 18, no. 10, pp. 1–22, 10 2024. [Online]. Available: <https://doi.org/10.1371/journal.pntd.0012599>
- [33] P.-S. Lin, W.-L. Liu, C.-D. Chen, T.-H. Wen, C.-H. Chen, L.-W. Chen, and Y.-H. Kung, "Micro-scale urbanization-based risk factors for dengue epidemics," *International Journal of Biometeorology*, vol. 68, no. 1, pp. 133–141, 2023. [Online]. Available: <https://doi.org/10.1007/s00484-023-02577-2>
- [34] D. Moukheiber, D. Restrepo, S. A. Cajas, M. P. A. Montoya, L. A. Celi, K.-T. Kuo, D. M. Lo'pez, L. Moukheiber, M. Moukheiber, S. Moukheiber, J. S. Osorio-Valencia, S. Purkayastha, A. R. Paddo, C. Wu, and P.-C. Kuo, "A multimodal framework for extraction and fusion of satellite images and public health data," *Scientific Data*, vol. 11, no. 1, 2024. [Online]. Available: <https://doi.org/10.1038/s41597-024-03366-1>
- [35] W. Rehman, M. Nasar-u Minallah, and I. Butt, "Spatial mapping of dengue fever prevalence and its association with geo-climatic factors in lahore, pakistan," *Environmental Monitoring and Assessment*, vol. 196, no. 9, 2024. [Online]. Available: <https://doi.org/10.1007/s10661-024-12967-7>
- [36] I. G. Conforme, W. Salinas Moran, M.-A. Quiroz-Martinez, and M.-D. Gomez-Rios, "Model for predicting dengue fever using machine learning in an ecuadorian context," in *International Conference on Applied Technologies*, M. Botto-Tobar, L. Lema Moreta, M. Zambrano Vizuete, S. Montes Leo'n, P. Torres-Carrion, and B. Durakovic, Eds. Cham: Springer Nature Switzerland, 2026, pp. 278–295.
- [37] M. A. Bhuiyan, M. R. S. Akash, R. Islam, S. I. Polash, and S. A. Khushbu, "Early dengue prediction in bangladesh: A comparative study with feature analysis, explainable artificial intelligence, and model optimization," *Journal of Tropical Medicine*, vol. 2025, p. 1709439, 2025.
- [38] M. S. Islam *et al.*, "Sociodemographic and clinical factors associated with dengue infection in bangladesh: a systematic review and meta-analysis," *BMC Infectious Diseases*, vol. 22, no. 1, p. 597, 2022.
- [39] X. Lu, S. Y. Teh, C. J. Tay, N. F. Abu Kassim, P. S. Fam, and E. Soewono, "Application of multiple linear regression model and long short-term memory with compartmental model to forecast dengue cases in selangor, malaysia based on climate variables," *Infectious Disease Modelling*, vol. 10, no. 1, pp. 240–256, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2468042724001222>
- [40] J.-H. Kim, A.-Y. Lim, and S. H. Kim, "Evaluating the effectiveness of dengue surveillance in the tropical and sub-tropical asian nations through dengue case data from travelers returning to the five western pacific countries and territories," *Travel Medicine and Infectious Disease*, vol. 64, p. 102802, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1477893925000080>
- [41] Y.-J. Xu, Y.-Y. Wu, W.-L. Yuan, and L.-C. Jiang, "Realization of anti-glare algorithm based on stagnant water reflection of automobile headlights on rainy day," in *2022 10th International Conference on Information Systems and Computing Technology (ISCTech)*, 2022, pp. 696–701.
- [42] S. J. Basha, S. N. S. M. Kumar, S. VS, S. Hasure, L. R, D. SR, and P. Rajalakshmi, "Cyber physical system for stagnant water detection and targeted spraying using uavs," in *2026 IEEE Applied Sensing Conference (APSCON)*, 2026, pp. 1–4.
- [43] W. Yu, M. Wu, and Z. Ha, "Research on a pearson-lstm-am-based water quality prediction model for freshwater aquaculture," *Journal of the World Aquaculture Society*, vol. 56, no. 4, p. e70041, 2025. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/jwas.70041>
- [44] C. Dias, M. Santos, and F. Portela, "A swot analysis of big data in healthcare," pp. 256–263, 2020. [Online]. Available: <https://doi.org/10.5220/0009390202560263>
- [45] A. Abu Al Aish, G. Jaradat, M. Alshugran, and I. Alodat, "Automating swot analysis using machine learning methods," *International Journal of Advances in Soft Computing and its Applications*, vol. 13, pp. 139–161, 07 2021.